

H07A 7

= 0

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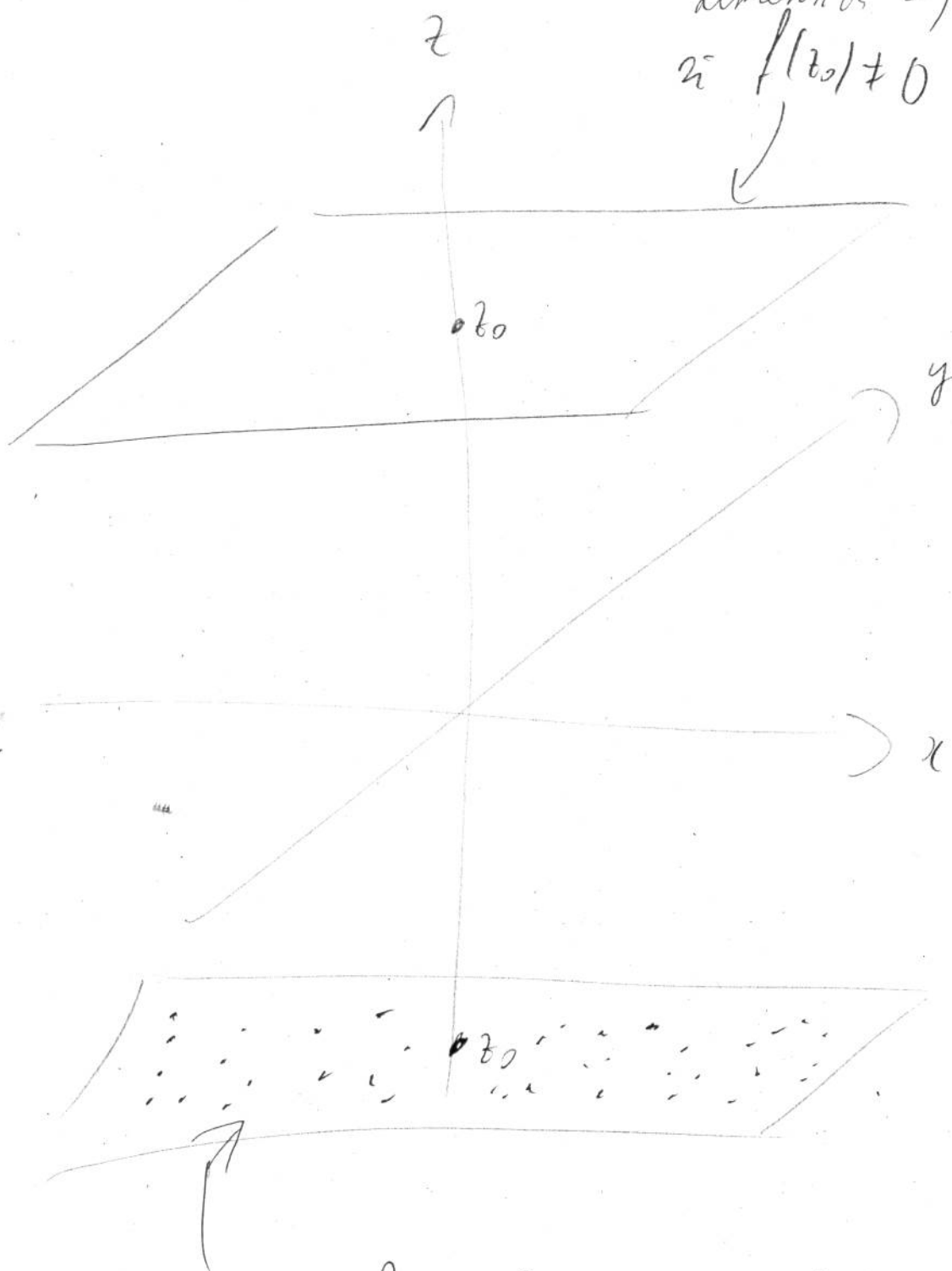
$$\begin{aligned} \textcircled{1} d/(\pi, \pi) &= \left(f(z) \cdot \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y}, f(z) \right) \wedge \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y} \\ &\quad \pm f(z) \cdot \left(f(z) \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y}, \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y} \right) \\ &\quad \pm \underbrace{\left(\frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y}, f(z) \right)}_{=0} \wedge \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y} \\ &= 0 \end{aligned}$$

$$\begin{aligned} h|_{\text{lineas}} \quad \# : T^* \mathbb{R}^3 &\rightarrow T^* \mathbb{R}^3 \\ dx &\mapsto f(z) \cdot \frac{\partial}{\partial y} \\ dy &\mapsto -f(z) \cdot \frac{\partial}{\partial x} \\ dz &\mapsto 0 \end{aligned}$$

$$\text{Entonces } \ker(\#_{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}) = \begin{cases} \ker \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) & \text{si } f(z) \neq 0 \\ \{0\} & \text{si } f(z) = 0. \end{cases}$$

$$\begin{aligned} \text{Entonces } \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} \in T^* \mathbb{R}^3 & \text{ la zona simplectica que pasa} \\ & \text{por } \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} \text{ es } \begin{cases} \mathbb{R}^2 \times \{z_0\} & \text{si } f(z_0) \neq 0 \\ \{(x_0, y_0, z_0)\} & \text{si } f(z_0) = 0 \end{cases} \end{aligned}$$

fuso de
 dimensión 2,
 $z \in f(z_0) \neq 0$



cada fuso es un punto,
 $z \in f(z_0) = 0$.

(2) Para toda aplicación lineal $A: V^* \rightarrow V$

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tenemos $\ker(A^*) = (\operatorname{Im} A)^0$, y o sea, $\ker(A^*) =$

$$\{z \in V^*: \langle v, A^* z \rangle = 0 \forall v \in V\} = \{z \in V^*: \langle Av, z \rangle = 0 \forall v \in V\}$$

$$= (\operatorname{Im} A)^0.$$

$\# : V^* \rightarrow V$ es antilineal, es decir, $\# = -\#^*$.

$$\text{Entonces } \ker(\#) = \ker(\#^*) = (\operatorname{Im} \#)^0.$$

(3) a) Sea $f \in C^\infty(U)$. Defina $F := \phi^* f - f \in C^\infty(D \times U)$.

Sea $p \in D$, así que $(p, \phi(p)) \in \text{gr}(\phi)$.

$$\begin{aligned} F(p, \phi(p)) &= (\phi^* f - f)(p, \phi(p)) \\ &= \underbrace{(\phi^* f)(p)}_{f(\phi(p))} - f(\phi(p)) = 0 \end{aligned}$$

b) Nota que si $f_i \in C^\infty(U)$, $g_i \in C^\infty(D)$ ($i = 1, 2$),

$$\text{entonces } \{g_1 + f_1, g_2 + f_2\} = \{g_1, g_2\}_{D \times U} - \{f_1, f_2\}_U.$$

1) Sean $f_1, f_2 \in C^1(\mathbb{R}^n)$, y defino $F_i := \phi^*(f_i) - f_i$ q.

como vect. Tenemos $\{F_1, F_2\}_{n \times n} =$

$$= \underbrace{\{\phi^*(f_1), \phi^*(f_2)\}_N}_{\leftarrow \phi \text{ es un morfismo de Poisson}} - \{f_1, f_2\}_N \in I_{\text{quib}}(\phi)$$

$$\phi^*(\{f_1, f_2\}_N)$$

c) Una función arbitraria en $I_{\text{quib}}(\phi)$ es una nueva fuente de términos de forma

$$G \cdot F, \text{ donde } G \in C^1(\mathbb{R}^n), \quad F = \phi^*(f) - f \text{ para } f \in C^1(\mathbb{R}^n)$$

Comprobamos que

$$\{G_1 F_1, G_2 F_2\}_{n \times n} = \underbrace{\{G_1 F_1, G_2\}_{n \times n}}_{\downarrow} \cdot F_2 + G_2 \cdot \underbrace{\{G_1 F_1, F_2\}_{n \times n}}_{\in I_{\text{quib}}(\phi)}$$

pertenece a $I_{\text{quib}}(\phi)$, ya que $F_2, \{F_1, F_2\}, F_1 \in I_{\text{quib}}(\phi)$.

Concluimos que $I_{\text{quib}}(\phi)$ es cerrado con respecto a $\{, \}_{n \times n}$.